

Complete NVRAM Analysis - First 13 Bytes (0x6000-0x600C)

ER3400 NVRAM Architecture

- Type:** ER3400 EAROM (Electrically Alterable ROM)
- Capacity:** 1024 × 4 bits
- Access:** Split into low nibble (0x60xx) and high nibble (0x62xx)
- Write Protocol:** Erase-before-write (erase at addr+0x400)

NVRAM Memory Layout

Configuration Data Structure (Bytes 0-10)

Template-driven configuration system using template at 0x0aa0: E8 E8 E8 FF F8 AF 80 80 80 80 80

NVRAM Byte	Physical Addresses	Template	Parameters	Function
0	0x6000/0x6200	0xE8	0-3	Serial Channel B + Base Config
1	0x6001/0x6201	0xE8	4-7	Serial Channel A Configuration
2	0x6002/0x6202	0xE8	8-11	Baud Rate Configuration
3	0x6003/0x6203	0xFF	12-19	Core System Control
4	0x6004/0x6204	0xF8	20-24	Extended Configuration
5	0x6005/0x6205	0xAF	25-30	Mixed Control Flags
6	0x6006/0x6206	0x80	31	Single Control Flag
7	0x6007/0x6207	0x80	32	Single Control Flag
8	0x6008/0x6208	0x80	33	Single Control Flag
9	0x6009/0x6209	0x80	34	Single Control Flag
10	0x600A/0x620A	0x80	35	Single Control Flag

Signature/Control Bytes (Outside Template System)

NVRAM Byte	Physical Addresses	Function	Value
11	0x600B/0x620B	Unknown/Reserved	?
12	0x600C/0x620C	Unknown/Reserved	?

Detailed Parameter Mapping

NVRAM Byte 0 (0x6000/0x6200) - Parameters 0-3

Template 0xE8: Extracts 4 bit fields

Param	RAM Address	Function	Code Usage
0	0x5f00	terminal_config_ram (base)	Referenced throughout system
1	0x5f01	z80sio_chb_wr4_clock_config	Channel B clock configuration
2	0x5f02	z80sio_chb_wr4_char_config	Channel B character format
3	0x5f03	Delay parameter	Used in serial timing (0x0982)

NVRAM Byte 1 (0x6001/0x6201) - Parameters 4-7

Template 0xE8: Extracts 4 bit fields

Param	RAM Address	Function	Code Usage
4	0x5f04	z80sio_cha_wr5_config	Channel A WR5 register
5	0x5f05	z80sio_cha_wr4_clock_config	Channel A clock configuration
6	0x5f06	z80sio_cha_wr4_char_config	Channel A character format
7	0x5f07	Display timing parameter	initialize_terminal_display_mode

NVRAM Byte 2 (0x6002/0x6202) - Parameters 8-11

Template 0xE8: Extracts 4 bit fields

Param	RAM Address	Function	Code Usage
8	0x5f08	baud_rate_mask_high	Baud rate configuration
9	0x5f09	baud_rate_current	Current baud rate setting
10	0x5f0a	baud_rate_mask_low	Baud rate configuration
11	0x5f0b	Unknown parameter	General configuration

NVRAM Byte 3 (0x6003/0x6203) - Parameters 12-19 ★ CRITICAL

Template 0xFF: Extracts 8 individual bits

Bit	Param	RAM Address	Function	Code Usage
7	12	0x5f0c	Unknown flag	General configuration
6	13	0x5f0d	keyboard_processing_disabled_flag	vsync_interrupt_handler
5	14	0x5f0e	display_mode_flag	setup_serial_channel_A_and_B
4	15	0x5f0f	Serial communication flag	Serial configuration
3	16	0x5f10	keyboard_data_available_flag	serial_start_transmission_if_ready
2	17	0x5f11	Unknown flag	General configuration
1	18	0x5f12	display_attribute_mode	Communication Mode V24/V11
0	19	0x5f13	keyboard_transmission_enable	Keyboard control

NVRAM Byte 4 (0x6004/0x6204) - Parameters 20-24

Template 0xF8: Extracts 5 bit fields

Param	RAM Address	Function	Code Usage
20	0x5f14	Unknown parameter	General configuration
21	0x5f15	Unknown parameter	General configuration
22	0x5f16	Unknown parameter	General configuration
23	0x5f17	baud_rate_backup	Backup baud rate storage
24	0x5f18	Unknown parameter	General configuration

NVRAM Bytes 5-10 (0x6005-0x600A) - Parameters 25-35

Template 0xAF/0x80: Various bit field extractions

Param Range	Function	Template
25-30	Mixed control flags	0xAF (6 parameters)
31	Single control flag	0x80
32	Single control flag	0x80
33	Single control flag	0x80
34	Single control flag	0x80
35	Single control flag	0x80

Note: Parameter 34 (0x5f22) is referenced in display code.

NVRAM Address Space (Outside Template System)

Signature Bytes (NVRAM Test Function)

Logical Addr	Physical Addr	Function	Expected Value
0x50	0x6050/0x6250	Signature byte 1	0x80
0x51	0x6051/0x6251	Signature byte 2	0x80

Configuration Data Range (Template System)

Logical Addr	Physical Addr	Function
0x52-0x5C	0x6052-0x605C	Configuration data (11 bytes)

Checksum/Control Range

Logical Addr	Physical Addr	Function	Notes
0x5D	0x605D/0x625D	Checksum accumulator	Expected: 0xAA
0x5E	0x605E/0x625E	Configuration end marker	Loop termination

Unknown/Reserved (Bytes 11-12)

NVRAM Byte	Physical Addr	Function	Status
11	0x600B/0x620B	Unknown/Reserved	No code references found
12	0x600C/0x620C	Unknown/Reserved	No code references found

Critical System Mode Control

For system_mode_state = 2 (Normal Mode):

Required: NVRAM byte 3 (0x6003/0x6203) = 0x08

- **Bit 3 SET:** keyboard_data_available_flag = 1
- **Bit 2 CLEAR:** Unknown flag = 0

Communication Mode V24/V11:

Control: NVRAM byte 3 bit 1 (parameter 18)

- **V24:** Bit 1 SET (0x02)
- **V11:** Bit 1 CLEAR (0x00)

ER3400 Write Protocol

Write Sequence for Each Nibble:

1. **READ** current value at 0x6xxx
2. **ERASE** at 0x6xxx + 0x400 (A10=1)
3. **READ** verify erase
4. **WRITE** new value at 0x6xxx (A10=0)
5. **READ** verify write

Example: Writing to NVRAM byte 3

Lower nibble: 0x6003 → Erase: 0x6403 → Write: 0x6003
Upper nibble: 0x6203 → Erase: 0x6603 → Write: 0x6203

Menu System Integration

Communication Switches Menu Impact:

- **Communication Mode:** Controls parameter 18 (NVRAM byte 3 bit 1)
- **Send Receive Mode:** Controls parameter 15 (NVRAM byte 3 bit 4)
- **Unknown options:** Control parameters 16-17 (NVRAM byte 3 bits 3-2)

Configuration Flow:

1. **Startup:** NVRAM → RAM extraction via template
2. **Runtime:** System uses RAM parameters (0x5f00-0x5f23)
3. **Menu:** Direct RAM editing
4. **Save:** RAM → NVRAM packing via template

Factory Reset Function

Location: factory_reset_nvram @ 0x3820

Process: Writes default values to all NVRAM locations

Range: Covers addresses 0x50 through approximately 0x6B

Trigger: Configuration menu option 6

This analysis provides complete traceability from NVRAM physical storage through template extraction to runtime usage for the first 13 NVRAM bytes.